

PhonoSec: AI-Powered Speech Emotion Recognition on Live Calls

Cybercriminals frequently use live phone calls to conduct scams, extortion, and misinformation, where victims express distress and uncertainty. Traditional monitoring systems analyze only the content of calls, lacking real-time emotional intelligence.

PhonoSec is an AI-powered Speech Emotion Recognition (SER) system designed for cybercrime police. It captures real-time audio, extracts voice features such as pitch and tone, and applies deep learning models to classify emotions like fear, distress, and deception. The system provides immediate alerts to law enforcement officers when high-risk conversations are detected.

Utilizing advanced machine learning, speech processing techniques, and secure cloud-based infrastructure, PhonoSec enhances cybercrime investigations by offering real-time emotional insights, improving fraud detection, and aiding victim support. The project aims to revolutionize law enforcement capabilities, making cybercrime response more proactive and efficient.

Keywords: Speech Emotion Recognition, Cybercrime Investigation, AI-powered Fraud Detection, Real-time Emotion Analysis, Cybersecurity.